Stefan Popovski

Skopje, Macedonia / College Park, MD, U.S.A | **US**: +1(240) 825 8267 **MKD:** +389-70-254-809
popovskistefan12@gmail.com https://www.linkedin.com/in/popovskis/

EDUCATION

University of Maryland, College Park — College Park, MD
Bachelor of Science — Expected May 2027
Major in **Mechanical Engineering**
In-Major GPA: 3.7/4.0
Honors & Awards

- A. James Clark School of Engineering Dean's List — all semesters.
- Pi Tau Sigma, UMD Chapter — Mechanical Engineering Honor Society.

WORK EXPERIENCE

A. James Clark School of Engineering – Smart Materials and Structures Research Center (SMSRC)
Undergraduate Research Assistant under Dr., Amr Baz — May 2026 – Jul 2026
Smart Piezoelectric Composite Tires for Ground Platforms with Energy Harvesting and Monitoring Capabilities

- Frequency Analysis of Fourier Transformations of input signals of Tire Deflections and Sound across different surface conditions and tire deflection.
- Data collection through custom Arduino circuits, or NI DAQ & **LabView**.
- Building robust LabView data collection loops.
- Prototype design and assembly, including piezoelectric sensor soldering, CAD design/printing/prototyping.
- MATLAB data post-processing, filtering, and analysis.

A. James Clark School of Engineering – Dept. of Mechanical Engineering — College Park, MD
Undergraduate Teaching Fellow – ENME350: Electronics and Instrumentation I — Jan 2026 - Present

- Held weekly office hours, assisting students with their lab reports, homework assignments and provided resources for exam preparation.
- Instructed and proctored the lab portion of the course, assisting students with constructing circuits and aiding students using oscilloscopes, and passive circuit elements.
- Checked off students for completing the in-lab projects, and collected data, and graded their respective In-Lab reports through Gradescope.

TSG Industry Service DOO, Skopje (North Macedonia) – DESFA S.A. (Greece) — Skopje, Macedonia
Engineering Intern — May 2025 – Aug 2025

- Assessed pipeline degradation through corrosion rate analysis to inform maintenance and failure-risk decisions for FPSO crude oil rig infrastructure (Trident Energy).
- Assisting senior engineering team in developing the Macedonia-Greece 68 km natural gas interconnector, providing a stable and more efficient flow of gas, using **AutoCAD** to analyze piping system and pull dimensions.
- Created a **Python-based** application which optimizes the best combination of 12-, 14- & 18-meter pipes to be fit, and minimizes the waste pipe drop-off at welded joints, minimizing the funding for the pipe order.
- FEA for failure/performance analysis of Pedestrian Overpass "Vlajko" in **Ansys Mechanical/Static Structural**, and modelling in **Ansys SpaceClaim**.

POLYESTERDAY (leading printing & marketing company in Macedonia) — Skopje, Macedonia
Practicing Intern — Jun 2021 – Aug 2021

- Assisting with research of the CRM (Customer Relationship Management) modules of the ERP (Enterprise Resource Planning)
- Analyzing and constructing detailed descriptions of the products listed on the online order program (www.print.mk)
- Online customer support for secondary-level customers of the online order program.
- Pre-press and elementary photo-editing.

UNDERGRADUATE RESEARCH

A. James Clark School of Engineering – Center for Risk and Reliability (CRR) — College Park, MD, U.S.A
Undergraduate Research Volunteer — Aug 2025 – Feb 2026

- Research volunteer alongside Dr. Yunfei Zhao and PhD candidate Mr., Nassar, in genetic algorithms, Markov processes, and Reinforcement Learning to dynamic probabilistic risk assessment (PRA) for nuclear reactor systems failure prediction and reliability modeling.
- Implementing PyTorch and Colabs to build a stochastic Genetic Algorithm using the Weibull distribution and Monte Carlo sampling for maintenance optimization and predicting MTBF (Mean Time Between Failures).

CERTIFCATIONS/LICENSES

NON-DESTRUCTIVE TESTING (NDT) Sector Cert Macedonia, 2025

- Liquid Penetrant Testing and Inspection – Level II Certification
  Certified in Accordance with **EN ISO 9712/SNT-TC-1A** Standards.
- Certified to conduct Liquid Penetrant inspections/testing (pre-cleaning, penetrant application, mid-cleaning, development, inspection, and report preparation), and writing testing/inspection instructions, equipment selection (penetrants, developers), and acceptance levels, according to **DIN EN ISO 3452-1/2/3**, and other standards for more specific testing scenarios (pipe welds, steel castings, boilers and pressure vessels).

UNIVERSITY & PERSONAL PROJECTS

Product Design and Manufacturing - Redesign Project: Bissell 3-in-1 Vacuum Cleaner March 2026 – May 2026

- SOLIDWORKS for CAD design of improved intake nozzle for performance improvement of budget vacuum cleaner.
- Project and Design Management regarding product price requirements, technical specification documentation, and mechanical properties of overall design (unit mass, reachability, surface, geometry, and application diversity).
- Presented ~150 slide redesign proposal presentation of redesign process, testing, application and results to a committee of 6 UMD Mech, Eng professors.

Product Design and Manufacturing: Thermal and Heat Study on DeWalt DCD701 Drill March 2026

- Thermistor calibration and data collection using Arduino circuit. Six thermistors were carefully placed on different locations on the drill casing and interior components.
- Presented ~80 slide presentation on our group's calculations of the drill's thermal efficiency, cooling methods, and overall heat transfer mechanisms.

NeuralSine: my first Neural Network

- Project, I dedicated to learning the architecture and structure of building Neural Network (forward pass, backpropagation, optimizers, loss and activation functions) algorithms.
- My program featured a data generator which plotted one period of sin(x), with a customizable gaussian noise addition for training and testing data.
- Built a neural network from scratch in NumPy to learn and predict the sine function, without relying on machine-learning frameworks.
- Tested different training methods, including SGD, momentum, and Adam, along with mini-batch training, learning-rate adjustments, and feature transformations; improved test error from ~0.4 to ~3e-5.
- Checked the model's calculations against numerical estimates to make sure the training process was working correctly and documented the full process and reasoning in a 6-page LaTeX report.
- Created a real-time Matplotlib visualization showing how the model's predictions and training performance changed throughout the learning process, with animated GIF export.

SignalAnalyzer: Simple DSP application and Fourier Analysis (FFT) June 2026

- Simple Digital Signal Processing program designed in Python for quickly adjustable Fourier analysis of raw output signals.
- Developed regarding my undergraduate research at the SMSRC, to cut-down on inefficient file/data switching and file mismatch in MATLAB.
- Implementing different signal processing and sampling algorithms (Rectangular, Hanning, Hamming, Blackman, and flat-top windows)
- Used and applied NumPy, Pandas, and Matplotlib.

Bearing Health Classifier Algorithm August 2026 – September 2026

- Built a logistic regression classifier in Python to predict bearing health (healthy class vs. faulty class) from raw accelerometer recordings sampled at 2560 Hz.
- Designed a signal processing and feature extraction pipeline that converts raw vibration signals into scaled feature inputs, including time-domain statistics such as RMS, kurtosis, and crest factor, and FFT for peak frequency detection.
- Trained the supervised model on labeled healthy and faulty datasets, then used it to classify unlabeled recordings of unknown bearing condition.
- Upgraded the model by using one-way ANOVA testing to rank extracted features by how well they separate healthy from faulty bearings.
- Built/trained an upgraded version using a trained Support Vector Machine (SVM) on the selected features and compared, which was able to classify between three classes (Healthy, Faulty Level I, and Fauly Level II)
- Used NumPy, Pandas, Matplotlib, and scikit-learn.

Car Cruise Controller (PI-Controller)

- Modelled a state-space representation of car dynamics/kinetics in Simulink and MATLAB, with a variable force input. This variable is controlled in real-time using a PI controller.
- Designed the block diagram of the PI controller, with inputs of the desired speed, and the actual speed of the car (state-space model), through a feedback channel to calculate the error signal.

- The error signal is sent through the respective proportional and integral gain blocks to control and output the force control, sent to the state-space model of a car, to adjust its acceleration and thus speed.
- Test-trials for testing various damping ratios, and natural frequencies through the mass-damper-spring analogy/equivalent of the PI Controller.

ADDITIONAL

Technical & General Skills

Communication, Critical Thinking, Team building, Organizational skills, SOLIDWORKS, Autodesk Inventor, Autodesk Fusion 360, MATLAB, C++, Python, NI LabView, Microsoft Office, Adobe Photoshop, Adobe Lightroom, PyTorch, TensorFlow, SciPy, Matplotlib, NI LabView, Microsoft Office, Microsoft Excel, Microsoft Word, CAD Slicing

Languages

Fluent in Macedonian & English. Elementary Proficiency in German & Serbian.

Relevant Coursework

ENME423 (Modern Climate Control and Building Energy Design/Analysis), ENME332 (Transfer Processes), ENES232 (Thermodynamics), ENME331 (Fluid Dynamics), MATH246 (Differential Eq. for Scientists & Engineers) ENES102/220 (Statics and Strength of Materials), ENME202 (Programming Fundamentals for Engineers), ENME272 (Introduction to CAD Design), **ENME462 (Control Algorithms) ,ENME382 (Introduction to Materials Engineering), ENME392 (Statistics and Probability for Engineers), ENME350/351 (Intro to Mechatronics and Analog/Digital Circuits), ENME361 (Mechanical Vibrations)**